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COURSE TITLE: AGILE DEVELOPMENT PROCESS AND DEVOPS

COURSE CODE: ISWE406L SLOT: C2+TC2

AGILE COURSE PROJECT

LMS APPLICATION

LMS APPLICATION – VERSION 1 (v1) IMPLEMENTATION

1. Introduction

The objective of this project is to develop a simple **Learning Management System (LMS)** that allows users to upload study materials. This version (v1) focuses on implementing the **Upload Notes feature**.

2. Tools and Technologies Used

- Visual Studio Code (IDE)
- Node.js (Backend runtime)
- Express.js (Web framework)
- Multer (File upload middleware)
- HTML (Frontend form for file upload)
- Web Browser (for testing)

3. Project Setup

3.1 Project Initialization

A new Node.js project was initialized using:

```
npm init -y
```

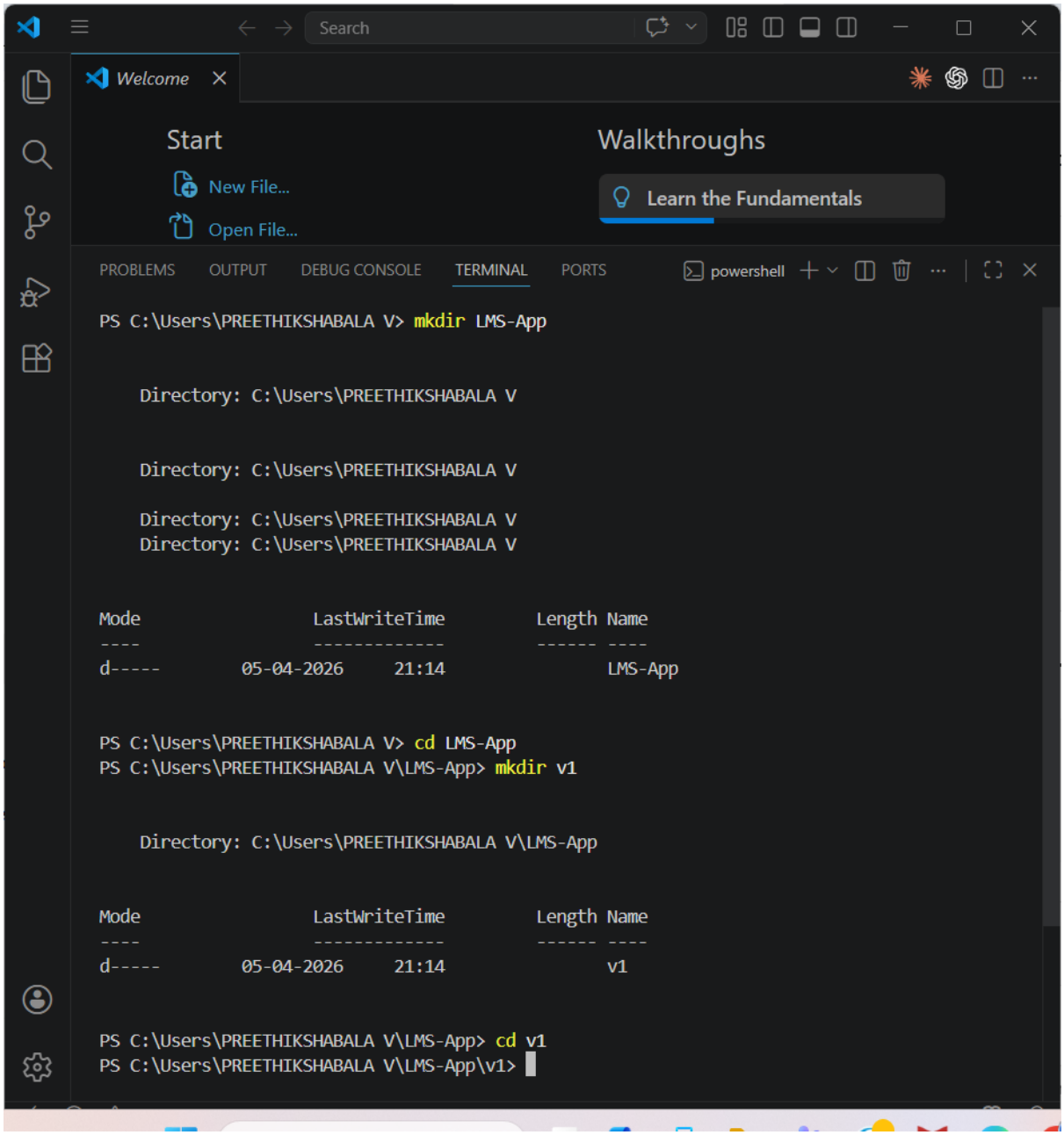
This created the package.json file.

3.2 Dependency Installation

Required dependencies were installed:

```
npm install express multer cors
```

- **Express** → Server framework
- **Multer** → Handles file uploads
- **CORS** → Enables cross-origin requests

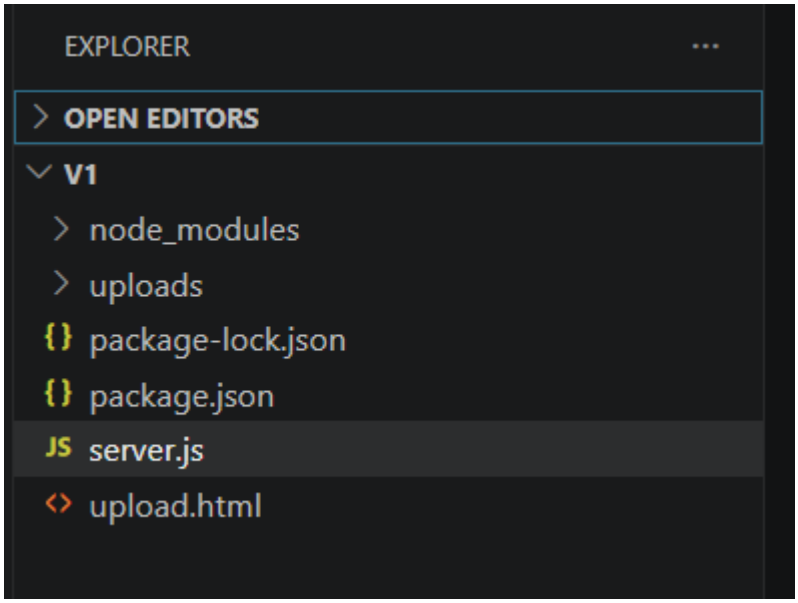

Search
Welcome
Start
New File...
Open File...
Walkthroughs
Learn the Fundamentals
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\PREETHIKSHABALA V> mkdir LMS-App
Directory: C:\Users\PREETHIKSHABALA V
Directory: C:\Users\PREETHIKSHABALA V
Directory: C:\Users\PREETHIKSHABALA V
Directory: C:\Users\PREETHIKSHABALA V
Mode LastWriteTime Length Name
---- ---------- ------ ----
d----- 05-04-2026 21:14 LMS-App
PS C:\Users\PREETHIKSHABALA V> cd LMS-App
PS C:\Users\PREETHIKSHABALA V\LMS-App> mkdir v1
Directory: C:\Users\PREETHIKSHABALA V\LMS-App
Mode LastWriteTime Length Name
---- ---------- ------ ----
d----- 05-04-2026 21:14 v1
PS C:\Users\PREETHIKSHABALA V\LMS-App> cd v1
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1>

```
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> npm init -y
Wrote to C:\Users\PREETHIKSHABALA V\LMS-App\v1\package.json:

{
  "name": "v1",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> npm install express multer cors

added 82 packages, and audited 83 packages in 6s

25 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1>
```

4. Project Structure

The project structure is as follows:



- server.js → Backend server
- uploads/ → Stores uploaded files
- upload.html → Frontend for uploading files

5. Backend Implementation

The backend was developed using Express.js with Multer for handling file uploads.

API Endpoint:

POST /upload

Functionality:

- Accepts file input from user
- Stores files in uploads/ folder
- Generates unique filenames using timestamps

6. Running the Application

The server was started using:

node server.js

Output:

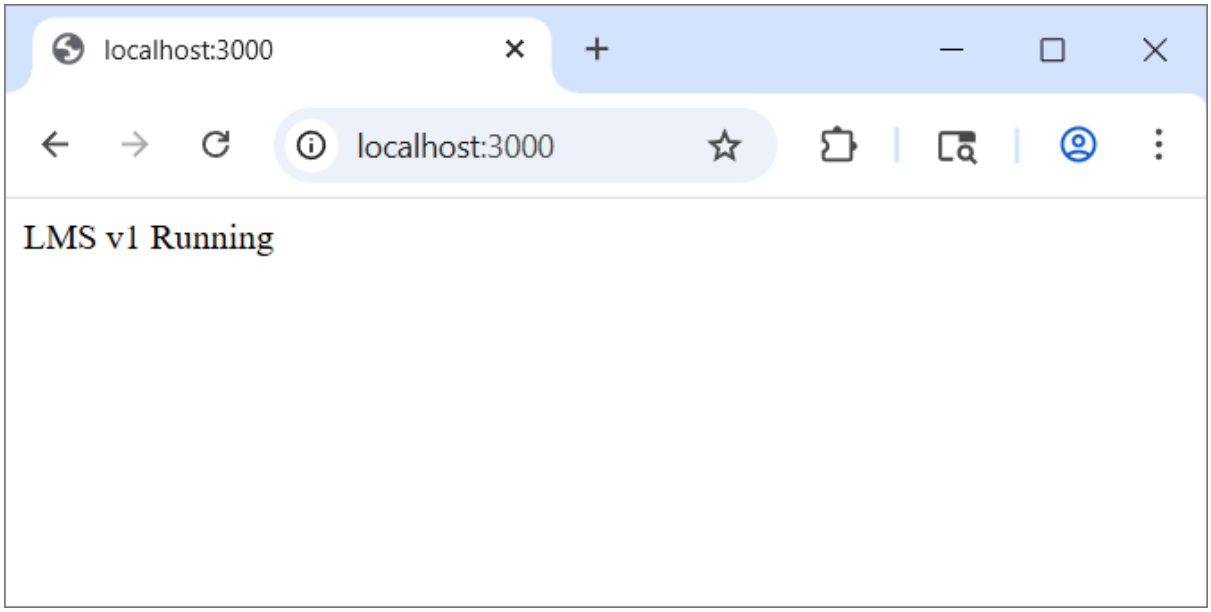
server.js

server.js > ...

```javascript
const express = require("express");
const multer = require("multer");
const cors = require("cors");

const app = express();
app.use(cors());

// storage config
const storage = multer.diskStorage({
    destination: (req, file, cb) => {
        cb(null, "uploads/");
    },
    filename: (req, file, cb) => {
        cb(null, Date.now() + "-" + file.originalname);
    }
});

const upload = multer({ storage: storage });

// upload API
app.post("/upload", upload.single("file"), (req, res) => {
    res.json({ message: "File uploaded successfully" });
});

// home route
app.get("/", (req, res) => {
    res.send("LMS v1 Running");
});

app.listen(3000, () => {
    console.log("Server running on port 3000");
});
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> node server.js
Server running on port 3000
```



7. File Upload Implementation (HTML Form)

A simple HTML interface was created to upload files:

- File input field
- Submit button
- Form uses multipart/form-data

This allows users to upload files directly from the browser.

```
server.js    upload.html
upload.html > html > body > form
1  <!DOCTYPE html>
2  <html>
3  <body>
4
5  <h2>Upload Notes</h2>
6
7  <form action="http://localhost:3000/upload" method="post" enctype="multipart/form-data">
8    <input type="file" name="file" />
9    <br><br>
10   <button type="submit">Upload</button>
11 </form>
12
13 </body>
14 </html>
```

8. Testing the Application

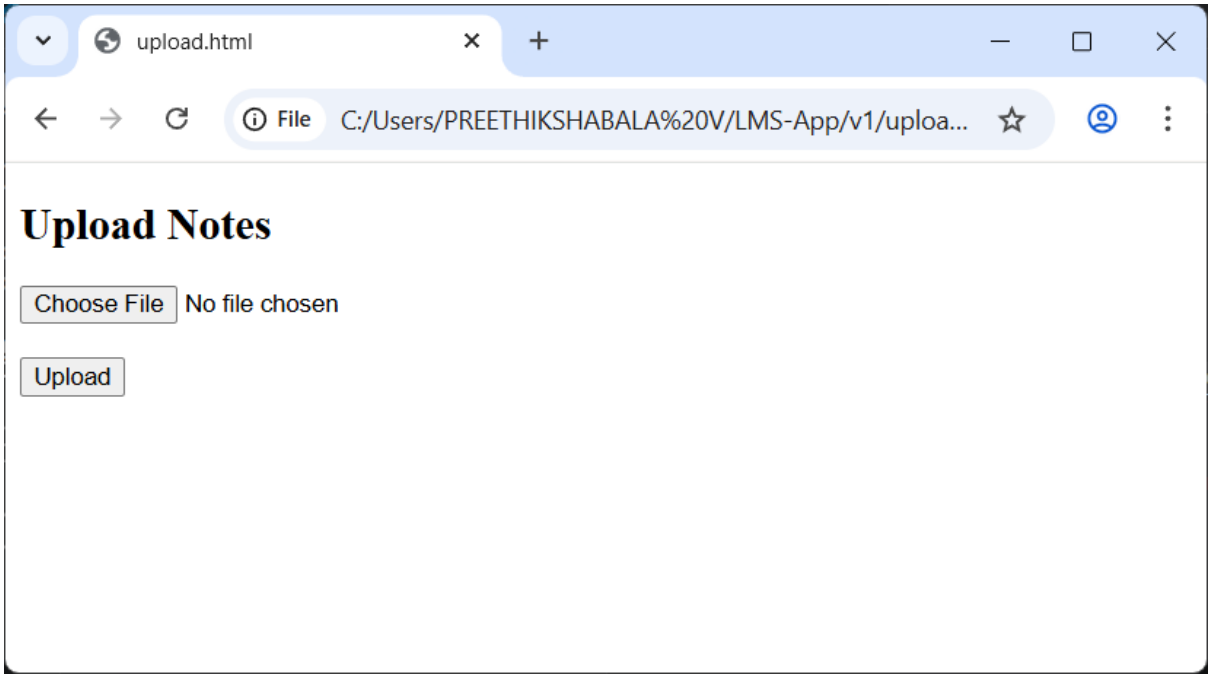
Steps:

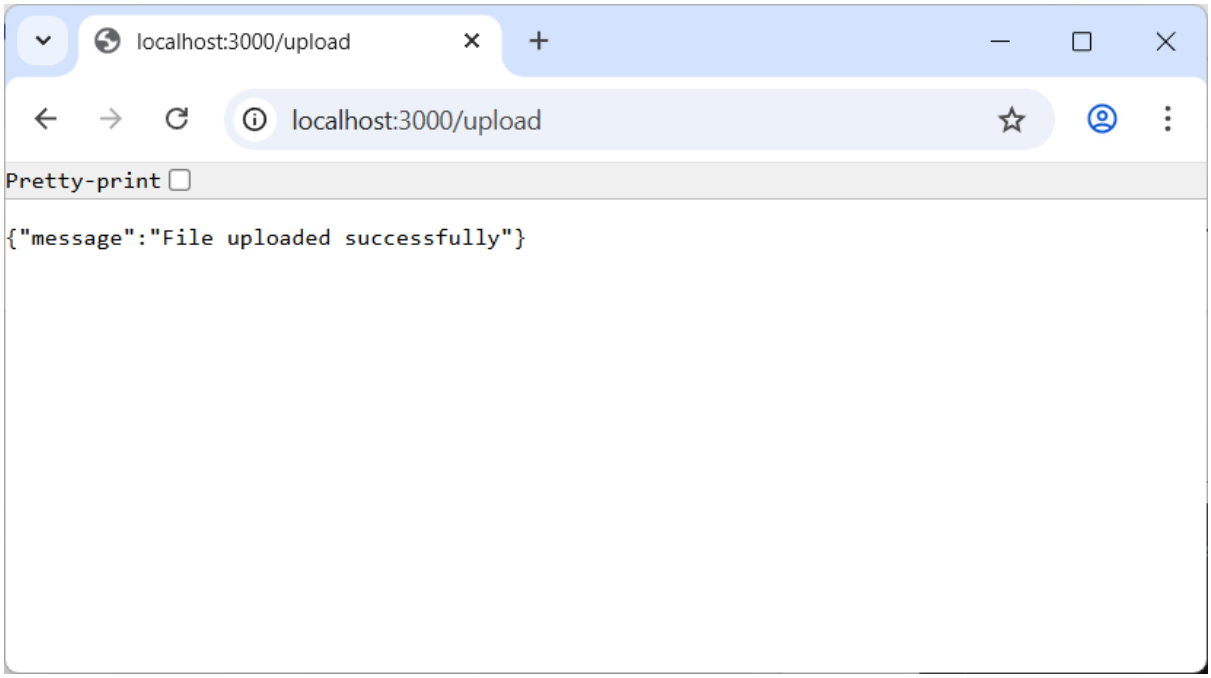
1. Open upload.html in browser

2. Select a file
3. Click Upload

Result:

- File successfully uploaded
- Stored in uploads/ folder

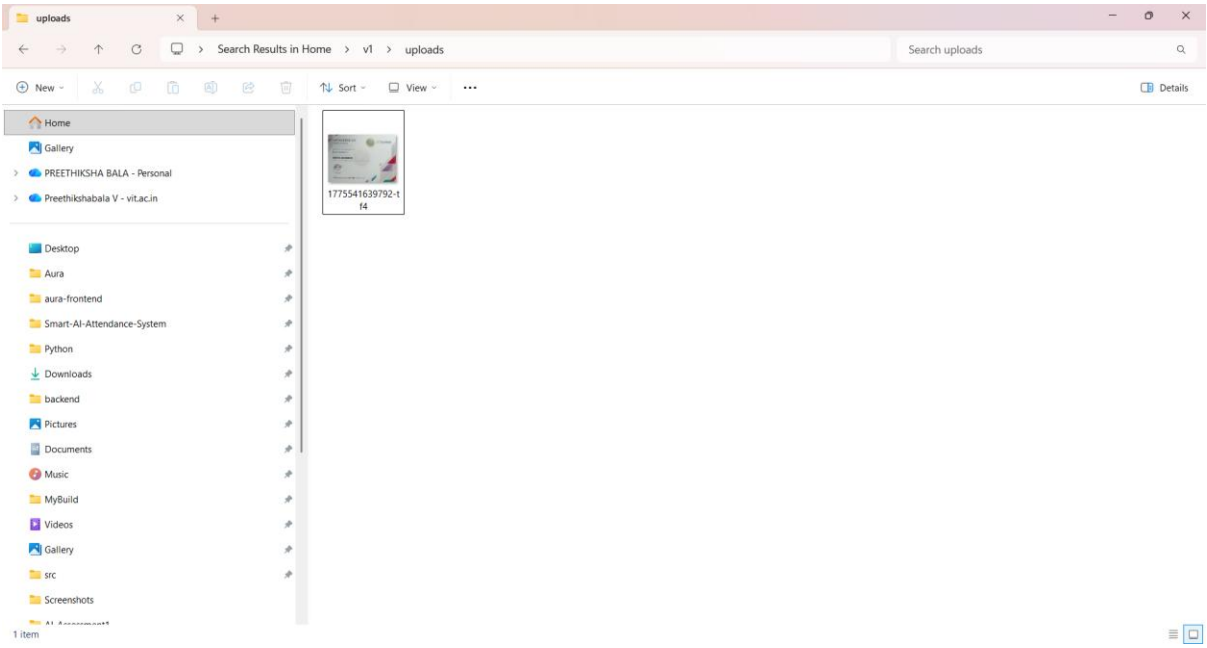




9. Output Verification

After uploading:

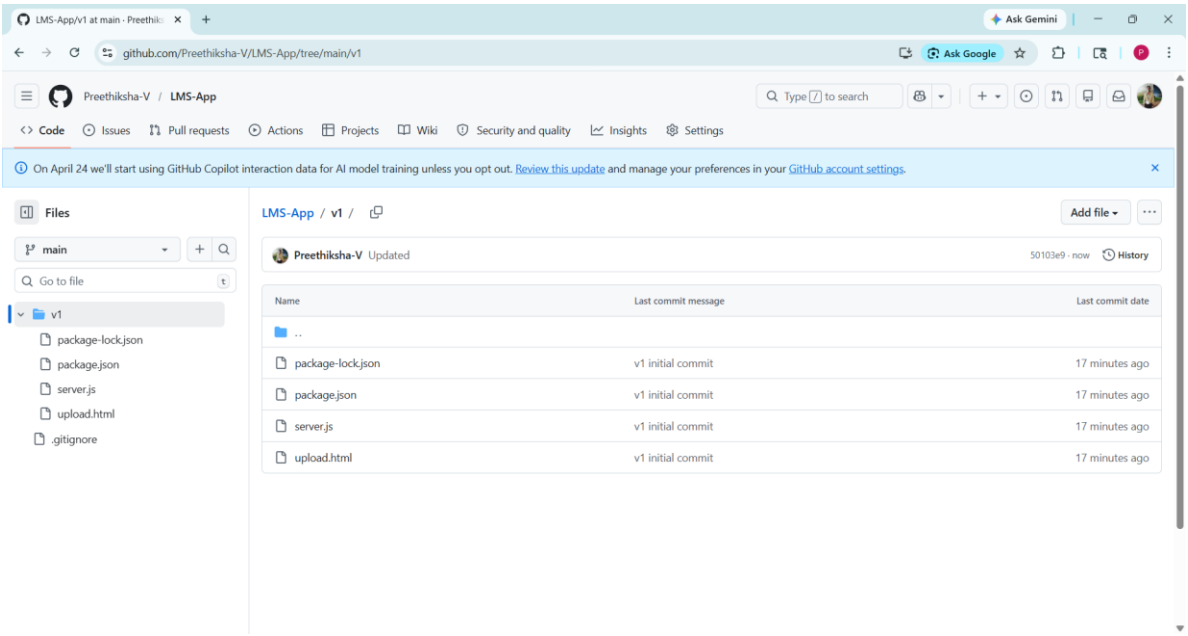
- The file appears inside the uploads/ directory
- Confirms successful backend integration



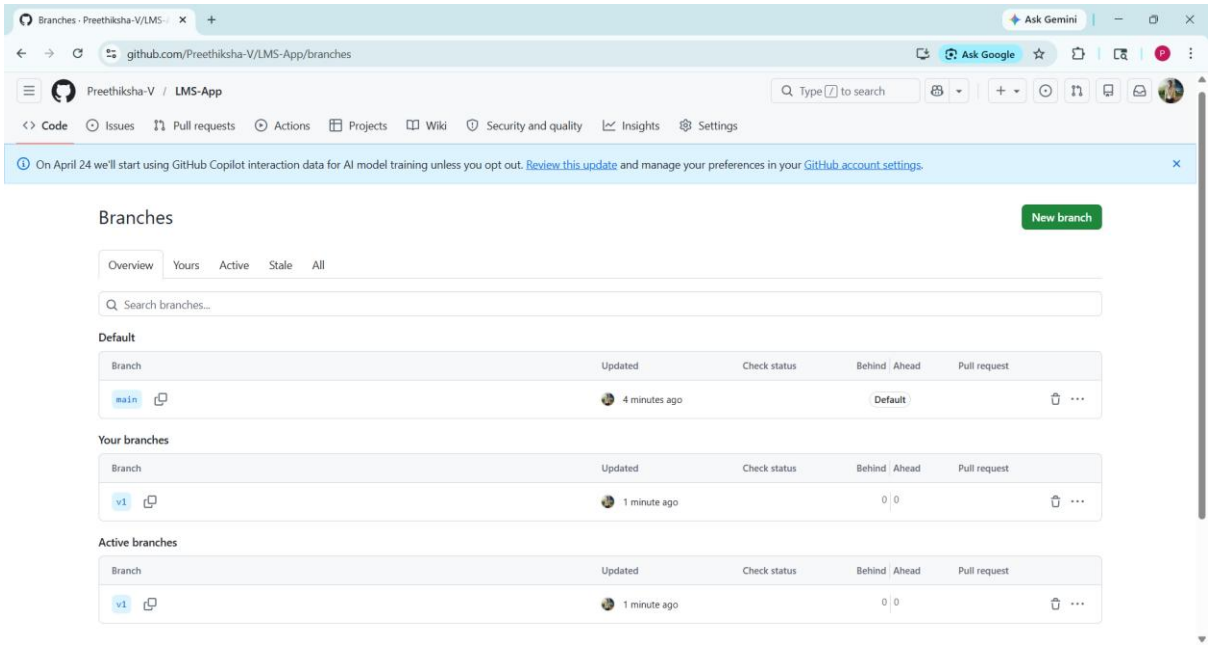
The LMS v1 application was successfully developed with:

- Functional backend server
- Working file upload feature
- Browser-based upload interface

This version forms the foundation for further enhancements such as viewing and downloading notes in future versions.



```
PS C:\Users\PREETHIKSHABALA V\LMS-App> git checkout -b v1
Switched to a new branch 'v1'
PS C:\Users\PREETHIKSHABALA V\LMS-App> git checkout -b v1
fatal: a branch named 'v1' already exists
PS C:\Users\PREETHIKSHABALA V\LMS-App> git push origin v1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'v1' on GitHub by visiting:
remote:      https://github.com/Preethiksha-V/LMS-App/pull/new/v1
remote:
To https://github.com/Preethiksha-V/LMS-App.git
 * [new branch]      v1 -> v1
PS C:\Users\PREETHIKSHABALA V\LMS-App>
```



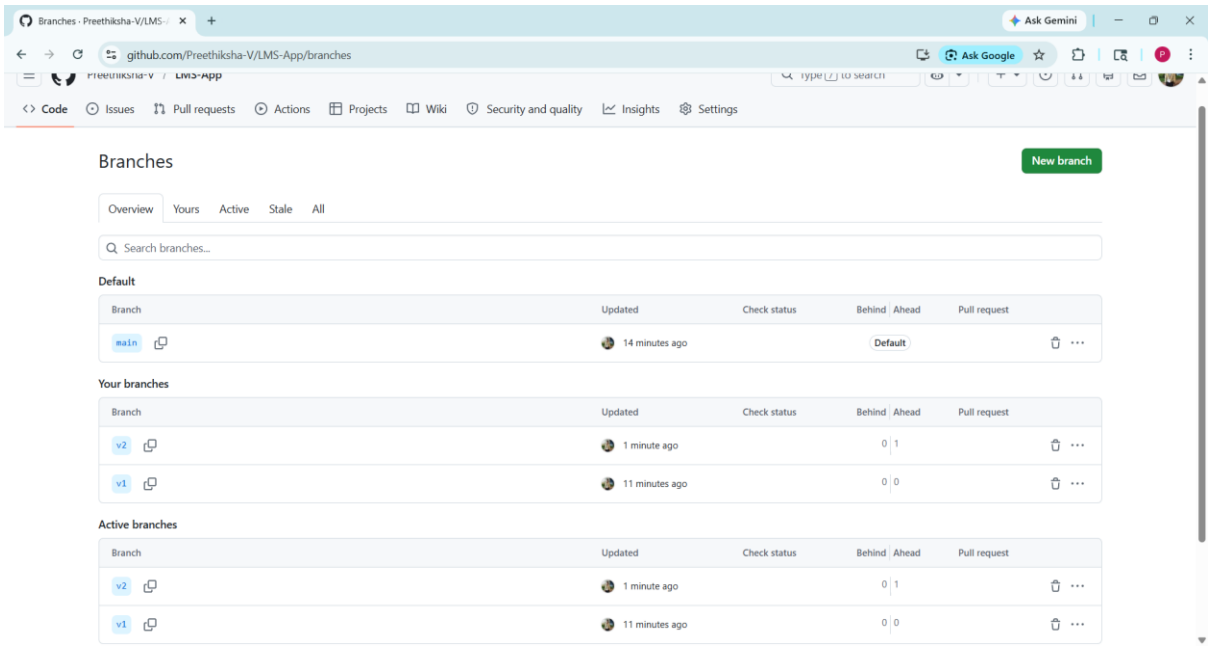

```
v1 > JS server.js > ...
});

app.listen(3000, () => {
    console.log("Server running on port 3000");
});

// view files API
app.get("/files", (req, res) => {
    const fs = require("fs");
    fs.readdir("uploads", (err, files) => {
        if (err) {
            return res.status(500).json({ error: "Unable to fetch files" });
        }
        res.json(files);
    });
});
```

localhost:3000/files

Pretty-print

["1775541639792-tf4.jpeg"]

```
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git add .
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git commit -m "v2 added view notes feature"
[v1 8eb8e34] v2 added view notes feature
 1 file changed, 10 insertions(+)
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git checkout -b v2
Switched to a new branch 'v2'
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git push origin v2
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 559 bytes | 186.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'v2' on GitHub by visiting:
remote:      https://github.com/Preethiksha-V/LMS-App/pull/new/v2
remote:
To https://github.com/Preethiksha-V/LMS-App.git
 * [new branch]      v2 -> v2
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1>
```



```
Welcome    .gitignore    server.js M
v1 > server.js > ...
// view files API
app.get("/files", (req, res) => {
    const fs = require("fs");
    fs.readdir("uploads", (err, files) => {
        if (err) {
            return res.status(500).json({ error: "Unable to fetch files" });
        }
        res.json(files);
    });
});
// download notes
app.get("/download/:filename", (req, res) => {
    const path = require("path");
    const file = path.join(__dirname, "uploads", req.params.filename);
    res.download(file);
});
```

```
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git add .
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git commit -m "v3 added download feature"
[v2 bb8245f] v3 added download feature
 1 file changed, 6 insertions(+)
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git checkout -b v3
Switched to a new branch 'v3'
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> git push origin v3
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 507 bytes | 253.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'v3' on GitHub by visiting:
remote:      https://github.com/Preethiksha-V/LMS-App/pull/new/v3
remote:
To https://github.com/Preethiksha-V/LMS-App.git
 * [new branch]      v3 -> v3
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1>
```

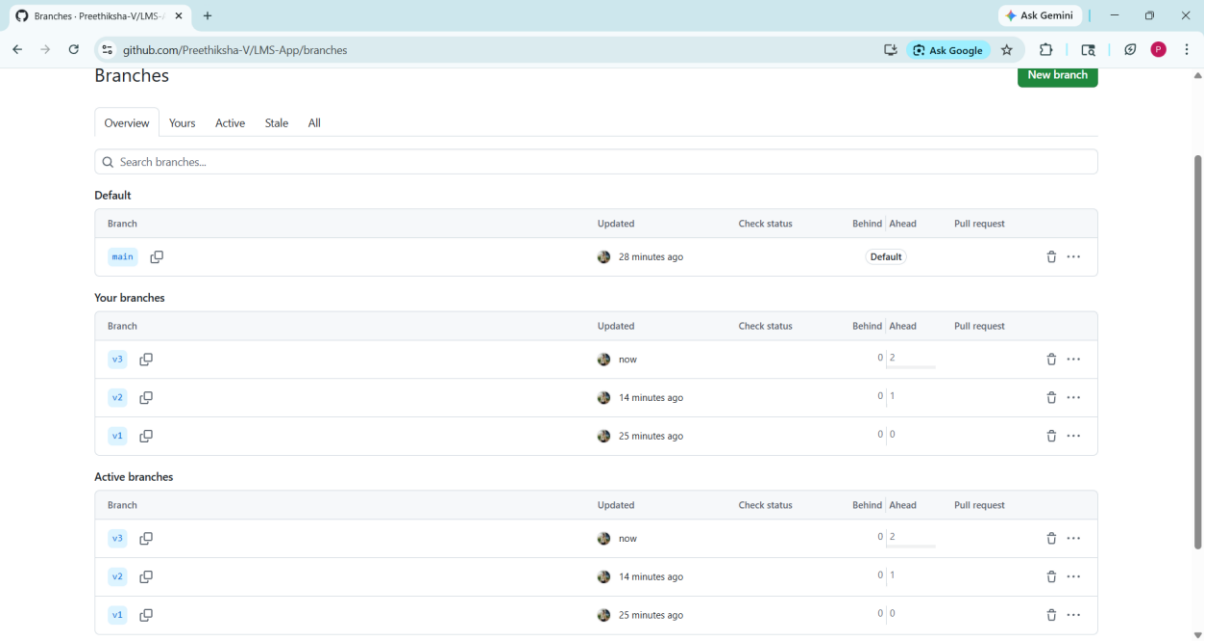
GITHUB

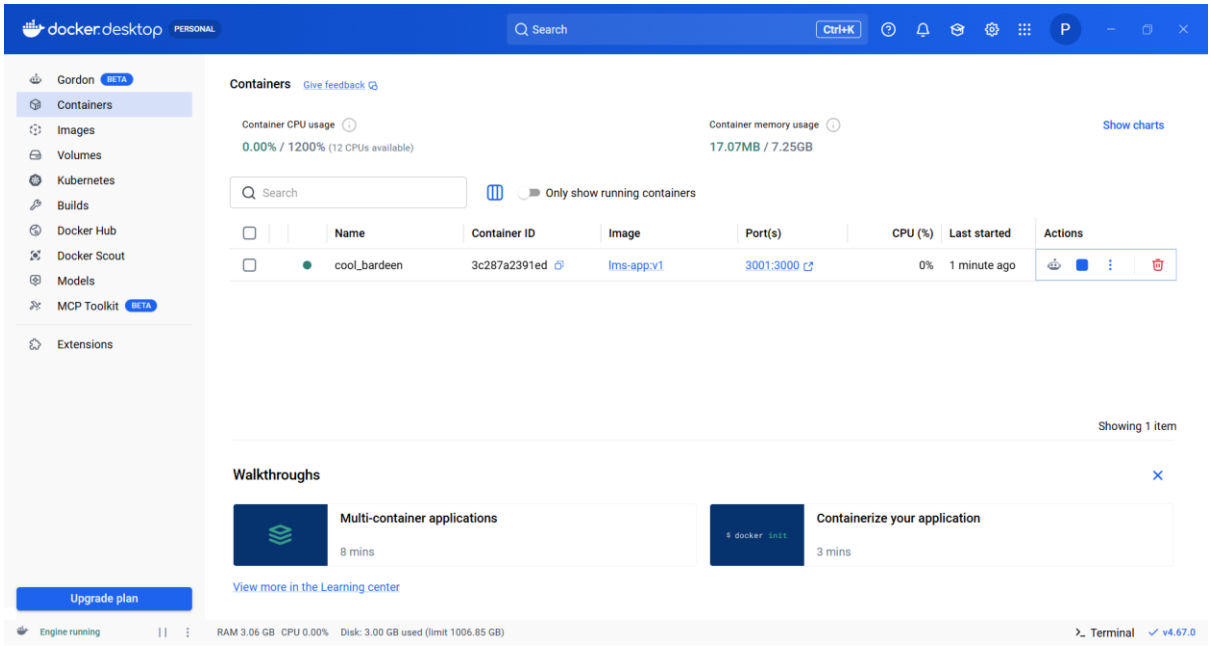
Git Push

Project code was pushed to GitHub repository successfully.

GitHub Repo View

Project files are visible in GitHub repository.

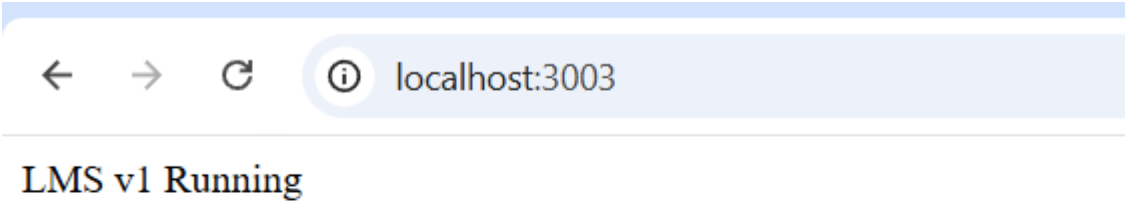


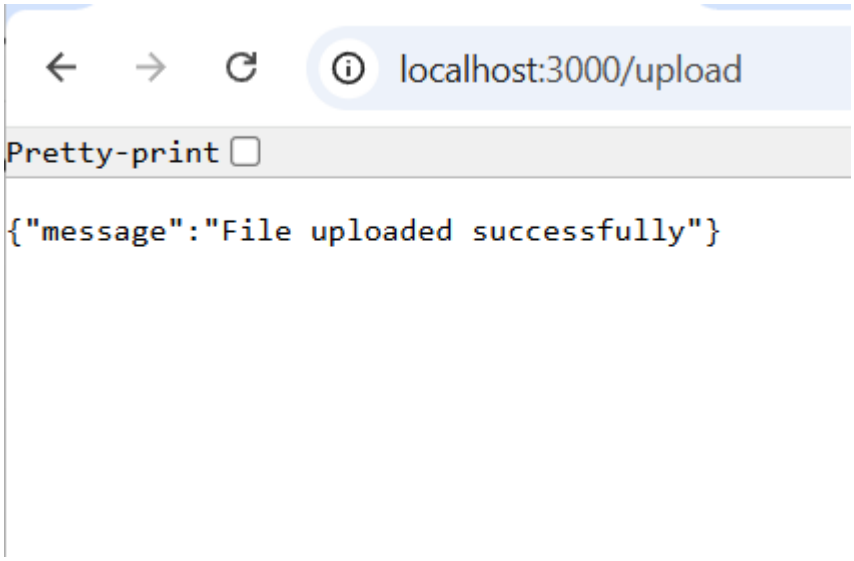


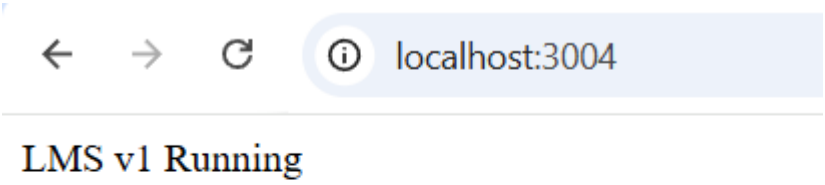
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                    docker - v1

PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> docker build -t lms-app:v1 .
 => => extracting sha256:461077a72fb7fe40d34a37d6a1958c4d16772d0dd77f572ec50a1fdc41a3754d
 => [2/5] WORKDIR /app
 => [3/5] COPY package*.json ./
 => [4/5] RUN npm install
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:4755b29dc1d12898be771d9193886f7fa983c6c33cc632e7b11d23ffaaf6aa38
 => => exporting config sha256:cbb44f253301910f0a666a0f2ba42a782b48cf4c7b3a7bb5530f23876f5bc4ff
 => => exporting attestation manifest sha256:7f99a2c0b32758b2a4df82ab03358e0d5f441813d1c50de5e574884f14e85ff2
 => => exporting manifest list sha256:4d60546d112b34f2bf8bde8ee4fd32f0769e3c53c5a1a27ad5f6881bdf0160a3
 => => naming to docker.io/library/lms-app:v1
 => => unpacking to docker.io/library/lms-app:v1
PS C:\Users\PREETHIKSHABALA V\LMS-App\v1> docker run -p 3001:3000 lms-app:v1
Server running on port 3000
```

localhost:3001

LMS v1 Running







DOCKER

Docker Build

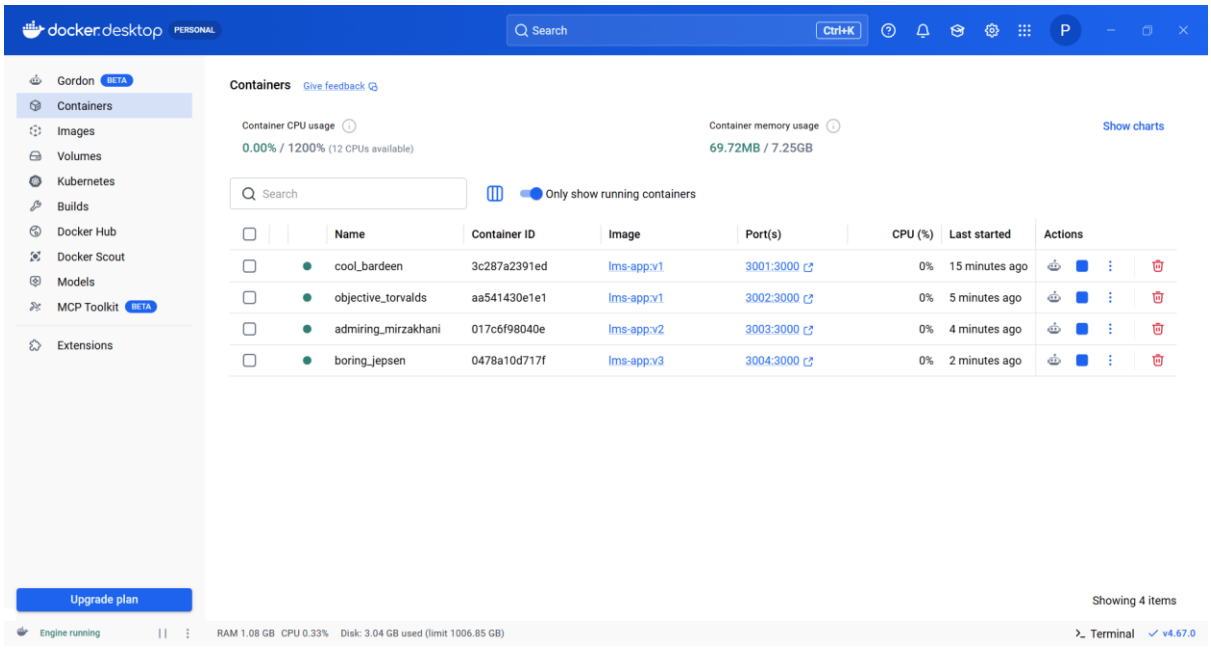
Docker image was built successfully for the LMS application.

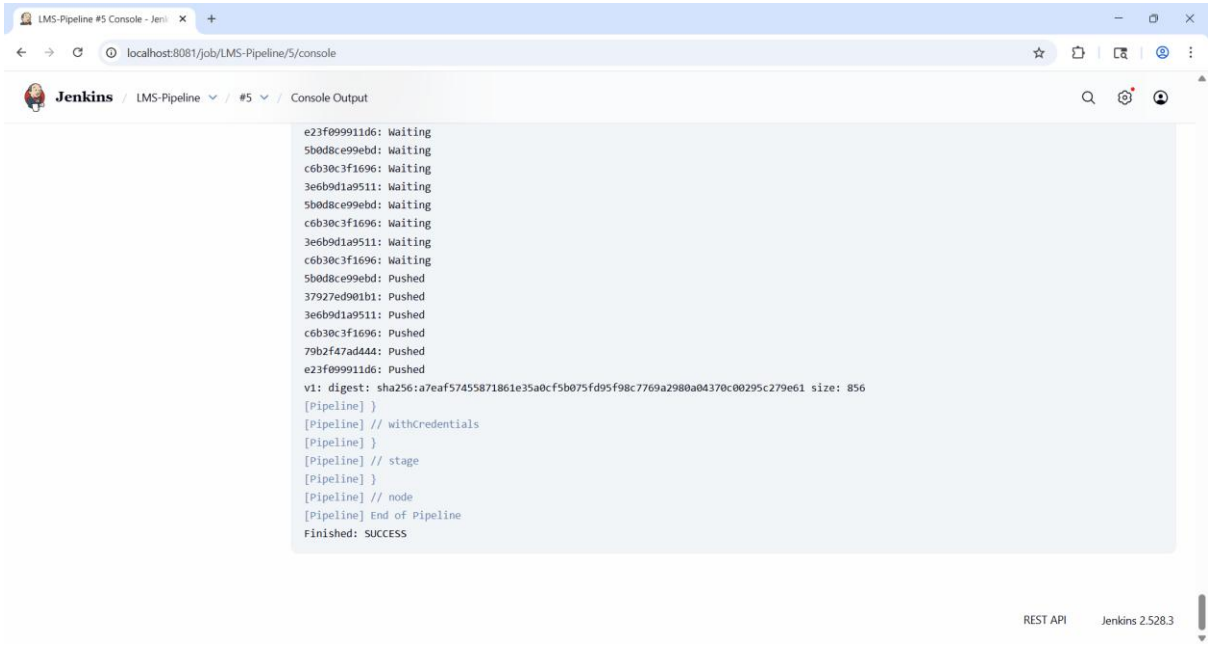
Docker Images List

Docker images for v1, v2, and v3 are created and available locally.

Docker Run

Docker container was executed and application is running on specified port.





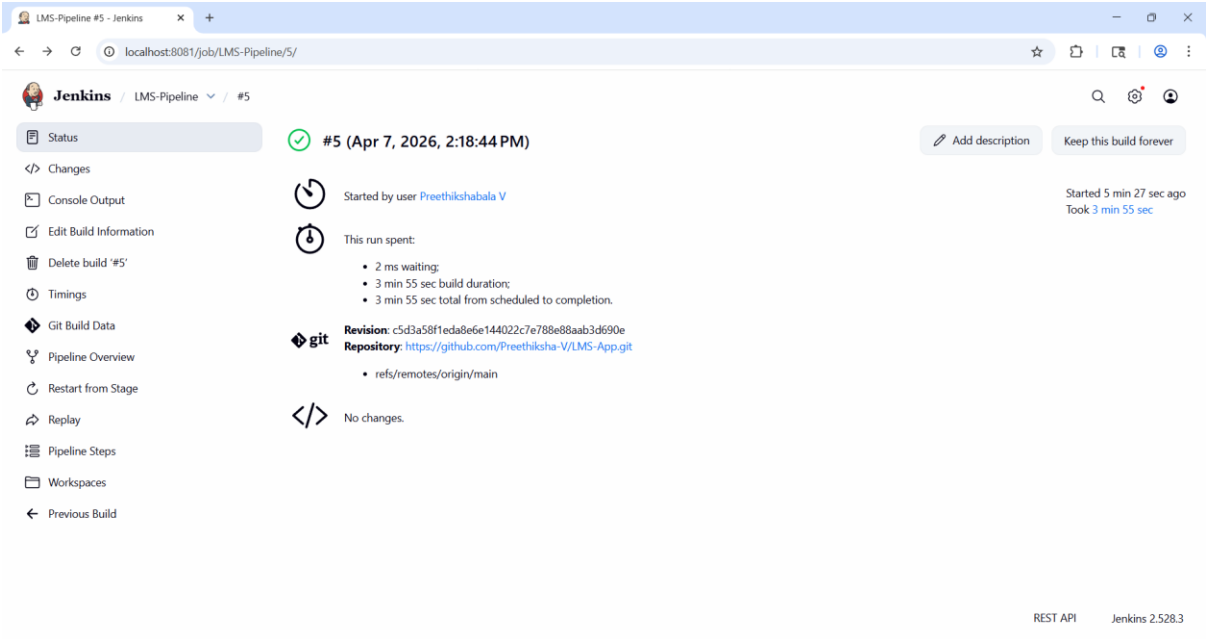
JENKINS

Jenkins Pipeline Start

Jenkins pipeline was triggered to automate build process.

Pipeline Success

Jenkins pipeline executed successfully including build and push stages.



KUBERNETES (v1)

kubectl get pods

Kubernetes pod for v1 is running successfully.

kubectl get services

Service is created to expose the application externally.

Port Forward

Port forwarding is used to access the application locally.

Browser Output v1

LMS v1 application is successfully running in browser.

v1-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lms-v1
spec:
  replicas: 1
  selector:
    matchLabels:
      app: lms-v1
  template:
    metadata:
      labels:
        app: lms-v1
    spec:
      containers:
      - name: lms-v1
        image: preethiksha/lms-app:v1
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 3000
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl apply -f v1-deployment.yaml
deployment.apps/lms-v1 created
```

v1-service.yaml

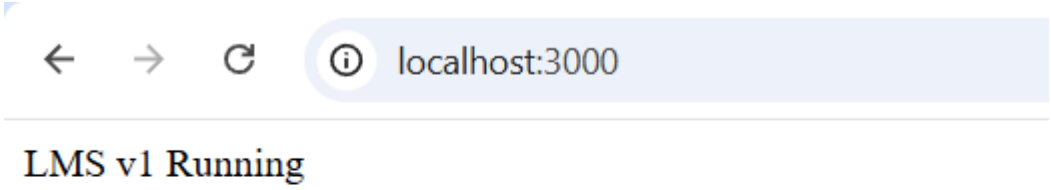
```yaml
apiVersion: v1
kind: Service
metadata:
  name: lms-v1-service
spec:
  type: NodePort
  selector:
    app: lms-v1
  ports:
    - port: 3000
      targetPort: 3000
      nodePort: 30007
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl apply -f v1-service.yaml
service/lms-v1-service created
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
lms-v1-794db5f564-r8cwh     1/1     Running   0          2m32s
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl get services
NAME             TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
kubernetes       ClusterIP   10.96.0.1      <none>        443/TCP          20m
lms-v1-service   NodePort    10.96.118.57   <none>        3000:30007/TCP   4m6s
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl port-forward service/lms-v1-service 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```



DEPLOY v2

v2 Pod Running
Kubernetes pod for v2 is successfully deployed and running.

Browser Output v2
LMS v2 application is running with updated functionality

v2-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lms-v2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: lms-v2
  template:
    metadata:
      labels:
        app: lms-v2
    spec:
      containers:
      - name: lms-v2
        image: preethiksha/lms-app:v2
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 3000
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl apply -f v2-deployment.yaml
deployment.apps/lms-v2 created
```

```yaml
apiVersion: v1
kind: Service
metadata:
  name: lms-v2-service
spec:
  type: NodePort
  selector:
    app: lms-v2
  ports:
    - port: 3000
      targetPort: 3000
      nodePort: 30008
```

```
C:\Users\PREETHIKSHABALA V\LMS-App\v1>kubectl apply -f v2-service.yaml
service/lms-v2-service created
```

v3

v3 Pod Running

Kubernetes pod for v3 is successfully deployed and running.

Browser Output v3

LMS v3 application is running successfully with final features.

```
C:\Users\PREETHIKSHABALA V\LMS-App\v3>kubectl apply -f v3-deployment.yaml
deployment.apps/lms-v3 created

C:\Users\PREETHIKSHABALA V\LMS-App\v3>kubectl apply -f v3-service.yaml
service/lms-v3-service created
```

All Pods Running

All three versions (v1, v2, v3) are running successfully in Kubernetes.

All Services

Services for all versions are created and accessible.